

Handwritten Digit Recognition

Project Report

1. Project Overview

This project demonstrates the use of Deep Learning and Convolutional Neural Networks (CNNs) to recognize handwritten digits. The model is trained using the MNIST dataset, which contains grayscale images of handwritten digits from 0 to 9. The objective is to build a classification model that can accurately identify handwritten digits from image data.

2. Technologies Used

Python, NumPy, Pandas, Matplotlib, TensorFlow, Keras, and Scikit-learn.

3. Dataset

The MNIST dataset was used for this project. It contains 60,000 training images and 10,000 testing images. Each image is a 28×28 grayscale image representing one of the digits 0 through 9.

4. Methodology and Steps Performed

1. Installed and imported the required Python libraries.
2. Loaded the MNIST handwritten digit dataset.
3. Visualized sample handwritten digit images.
4. Normalized pixel values from the range 0–255 to 0–1.
5. Reshaped images to include a channel dimension for CNN input.
6. Built a Convolutional Neural Network.
7. Compiled the model using the Adam optimizer and sparse categorical cross-entropy loss.
8. Trained the model for 5 epochs with a batch size of 128 and a validation split of 0.1.
9. Evaluated the trained model on unseen test data.
10. Generated training/validation accuracy and loss plots.
11. Generated sample predictions.
12. Generated a classification report and confusion matrix.
13. Created a model performance metrics table.
14. Saved the trained model as `handwritten_digit_cnn.keras`.

5. CNN Model Architecture

- Conv2D: 32 filters, 3×3 kernel, ReLU activation
- MaxPooling2D: 2×2 pooling
- Conv2D: 64 filters, 3×3 kernel, ReLU activation

- MaxPooling2D: 2×2 pooling
- Flatten layer
- Dense layer: 128 neurons with ReLU activation
- Dropout layer: rate 0.3
- Output layer: 10 neurons with Softmax activation

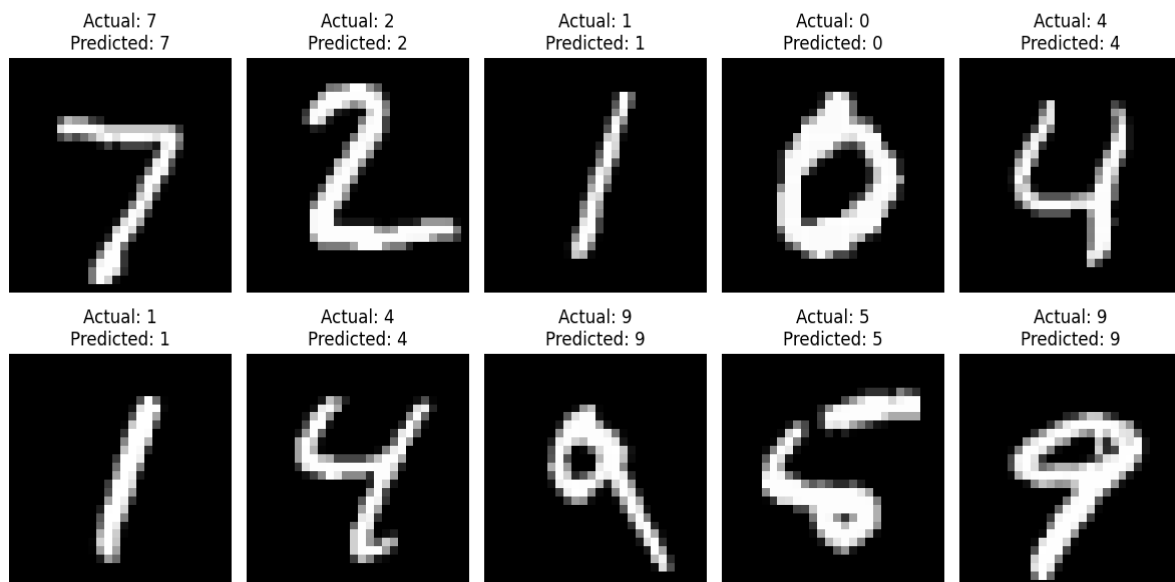
6. Training Results and Performance Metrics

Metric	Value
Training Accuracy	98.85%
Validation Accuracy	98.82%
Test Accuracy	98.71%
Test Loss	0.0374

The final training accuracy was 98.85%, the final validation accuracy was 98.82%, and the test accuracy was 98.71%. The test loss was 0.0374.

7. Sample Predictions

The model was tested on sample unseen MNIST images. The figure below compares actual and predicted digit labels.



8. Classification Report

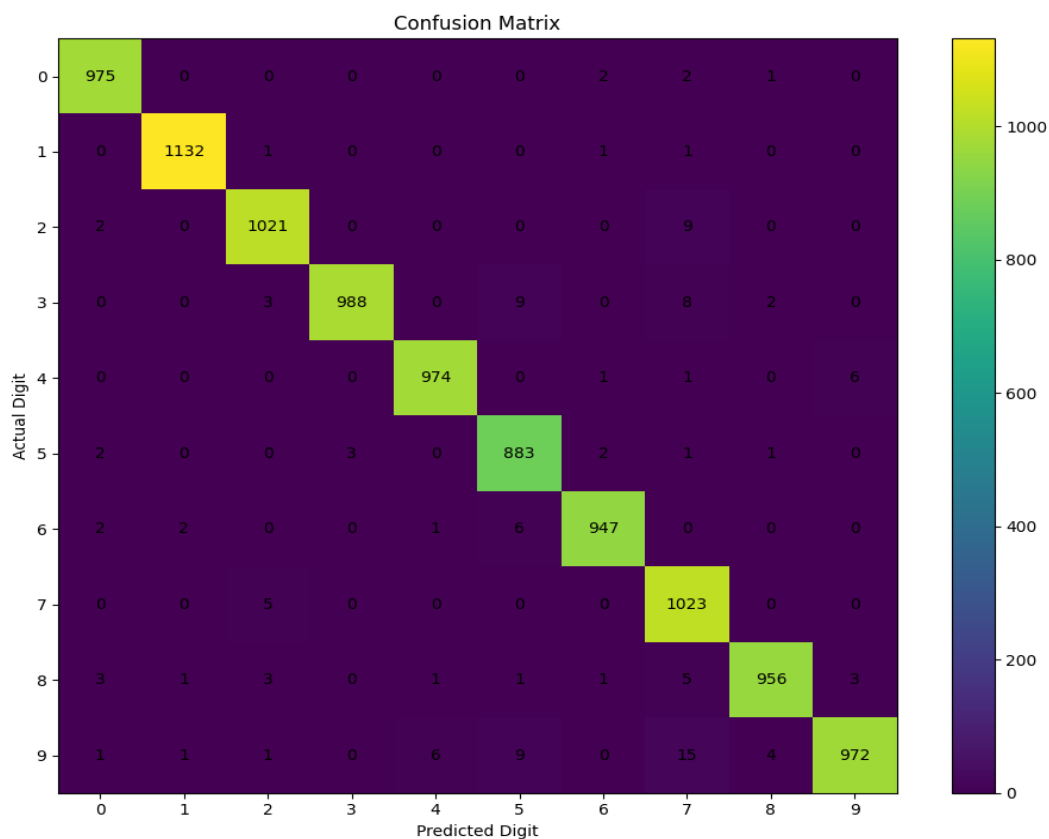
Digit	Precision	Recall	F1-Score
0	0.99	0.99	0.99
1	1.00	1.00	1.00
2	0.99	0.99	0.99

3	1.00	0.98	0.99
4	0.99	0.99	0.99
5	0.97	0.99	0.98
6	0.99	0.99	0.99
7	0.96	1.00	0.98
8	0.99	0.98	0.99
9	0.99	0.96	0.98

The overall accuracy reported for the 10,000 test images was approximately 0.99, with macro and weighted averages of approximately 0.99.

9. Confusion Matrix

A confusion matrix was generated to compare actual digit classes with predicted digit classes and to examine classification errors.



10. Conclusion

This project demonstrates the practical implementation of Deep Learning for handwritten digit recognition. A Convolutional Neural Network was built and trained using the MNIST dataset to classify handwritten digits from 0 to 9. The trained model achieved a test accuracy of 98.71% on unseen test data. Its performance was analyzed using training and validation metrics, sample predictions, a classification report, and a confusion matrix. The project demonstrates how CNNs can learn image features and perform accurate image classification.

11. Deliverables

Training notebook: Handwritten_Digit_Recognition.ipynb

Saved model: handwritten_digit_cnn.keras

Project report: Handwritten_Digit_Recognition_Report